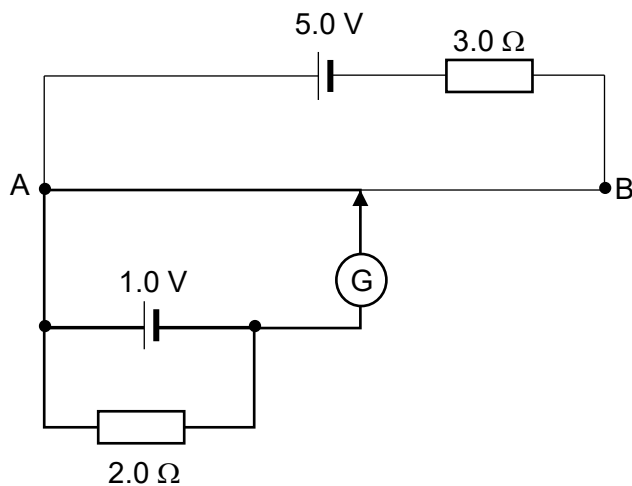


19. A potentiometer circuit using a 1.0 m resistance wire of resistance $1.0\ \Omega$ placed across AB is shown below. Both cells are ideal:

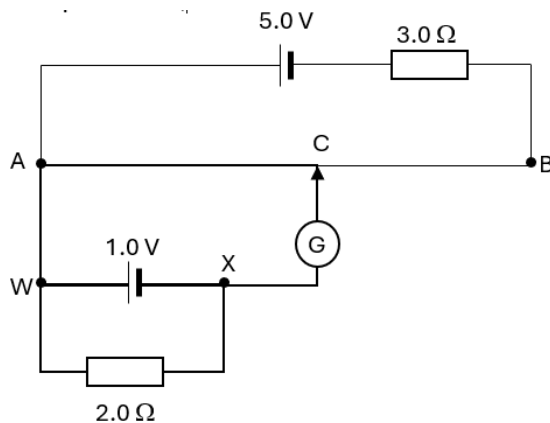


What is the balance length required for the galvanometer to show a null deflection?

- A** 20 cm **B** 50 cm **C** 60 cm **D** 80 cm

Ans: D

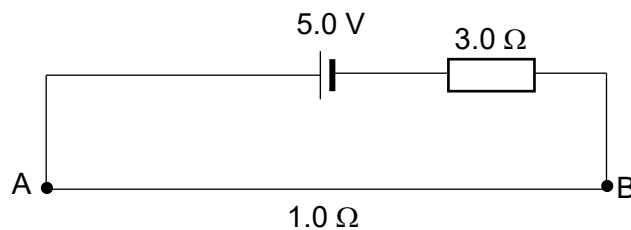
Let balance point be C;



At balance, $V_{AC} = V_{WX} = 1.0\ \text{V}$

So what length of AC will give 1.0 V?

Find V_{AB} first. Look at the top circuit and use potential divider principle:



$$\frac{V_{AB}}{\text{e.m.f}} = \frac{R_{AB}}{\text{total resistance}}$$

$$\frac{V_{AB}}{5.0} = \frac{1.0}{1.0 + 3.0}$$

$$V_{AB} = 1.25 \text{ V}$$

p.d. across AC is proportional to length AC.

$$\frac{AC}{AB} = \frac{V_{AC}}{V_{AB}}$$

$$\frac{AC}{1.0} = \frac{1.0}{1.25}$$

$$AC = 0.80 \text{ m}$$